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CSE427

Section 06

Assignment 2

TOPIC NAME : _____

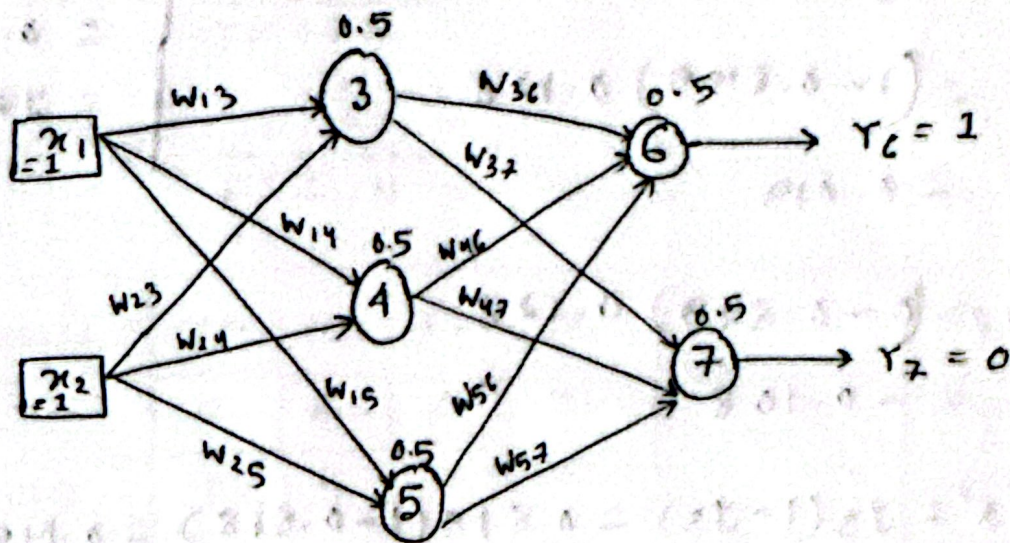
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②

$\eta = 0.1$



Sigmoid, $\sigma(z) = \frac{1}{1+e^{-z}}$

$W = 0.5$
 $b = 0.5$
 $\eta = 0.1$

$$y_3 = \sigma(w_{13}x_1 + w_{23}x_2 + b_3)$$

$$= \sigma(0.5 \times 1 + 0.5 \times 1 + 0.5)$$

$$= \sigma(1.5) = 0.818$$

$$y_4 = 0.818$$

$$y_5 = 0.818$$

$$y_6 = \sigma(w_{36}y_3 + w_{46}y_4 + w_{56}y_5 + b_6)$$

$$= \sigma(0.5(0.818 + 0.818 + 0.818) + 0.5)$$

$$= \sigma(1.727)$$

$$= 0.849$$

$$y_7 = 0.849$$

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$$\delta_6 = (1 - 0.849) y_6'$$

$$= (1 - 0.849) 0.128$$

$$= 0.019$$

$$y_6' = y_6(1 - y_6)$$

$$= 0.128$$

$$= y_7'$$

$$\delta_7 = (1 - 0.849) 0.128$$

$$= -0.108$$

$$y_3' = y_3(1 - y_3) = 0.818(1 - 0.818) = 0.149$$

$$= y_4' = y_5'$$

$$\delta_3 = (w_{36} \delta_6 + w_{37} \delta_7) y_3'$$

$$= (0.5 \times 0.019 + 0.5(-0.108)) \times 0.149$$

$$= -6.63 \times 10^{-3} = \delta_4 = \delta_5$$

$$w_{13} = w_{13} + \eta \delta_3$$

$$= 0.5 + 0.1 \times 1 (-6.63 \times 10^{-3})$$

$$= 0.499 = w_{23} = w_{14} = w_{24} = w_{25}$$

$$b_3 = b_3 + \eta \delta_3 = 0.5 + 0.1 (-6.63 \times 10^{-3})$$

$$= 0.499$$

$$b_4 = 0.499$$

$$b_5 = 0.499$$

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(3)

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

Threshold	TP	FN	FP	TN	TPR	FPR
1	0	6	0	4	0.00	0.00
0.975	1	5	0	4	0.17	0.00
0.975	1	5	0	4	0.17	0.00
0.965	2	4	0	4	0.33	0.00
0.95	3	3	0	4	0.50	0.00
0.87	3	3	1	3	0.50	0.25
0.75	4	2	1	3	0.67	0.25
0.50	4	2	1	3	0.67	0.25
0.20	6	0	1	3	1.00	0.25
0.15	6	0	2	2	1.00	0.50
0.10	6	0	3	1	1.00	0.75
0.05	6	0	4	0	1.00	1.00
0.00	6	0	4	0	1.00	1.00

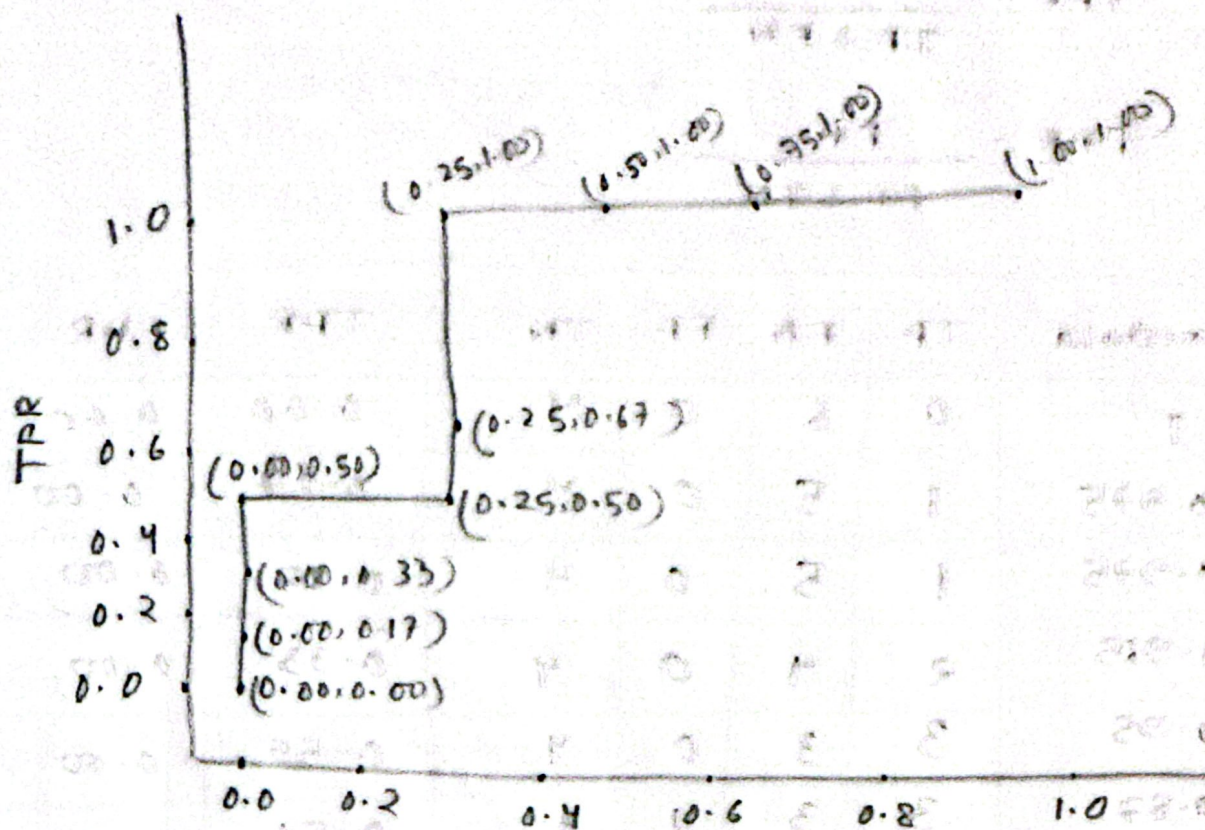
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100% TPR

Threshold $\rightarrow 0.20$

0% FPR

probability Threshold $> 1.00, 0.975, 0.965, 0.95$

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④

For class A:

$$\text{False positive} = 50 + 200 = 250$$

$$\text{False Negative} = 100 + 100 = 200$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{800}{800 + 200} = 0.8 = \text{Recall}$$

$$\text{Specificity} = \frac{TN}{FP + TN} = \frac{900 + 50 + 100 + 700}{(900 + 50 + 100 + 700) + (200 + 50)} = 0.875$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{800}{800 + 250} = 0.762$$

For class B

$$TP = 900$$

$$FP = 100 + 100 = 200$$

$$FN = 50 + 50 = 100$$

$$TN = 700 + 100 + 800 + 200 = 1800$$

$$\text{Sensitivity} = \text{Recall} = \frac{TP}{TP + FN} = \frac{900}{900 + 100} = 0.9$$

$$\text{Specificity} = \frac{TN}{FP + TN} = \frac{1800}{200 + 1800} = 0.9$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{900}{900 + 200} = 0.9$$

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For class c

$$TP = 700$$

$$FP = 100 + 50 = 150$$

$$FN = 200 + 100 = 300$$

$$TN = 3000 - (700 + 150 + 300) = 1850$$

$$\text{Sensitivity} = \text{Recall} = \frac{TP}{TP + FN} = \frac{700}{700 + 300} = 0.7$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{1850}{1850 + 150} = 0.925$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{700}{700 + 150} = 0.824$$

⑨

$$E_{we} = [1.87, 0.09]$$

$$E_{no} = [-0.78, 0.27]$$

$$PE(0,0) = \sin\left(\frac{0}{10000^{0.2}}\right) = 0$$

$$PE(0,1) = \cos\left(\frac{0}{10000^{0.2}}\right) = 1$$

$$PE(1,0) = \sin\left(\frac{1}{10000^{0.2}}\right) = 0.84$$

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$$PE(1,1) = \cos\left(\frac{1}{10000^{0.12}}\right) = 0.54$$

$$pos_{we} = [0, 1]$$

$$pos_{g0} = [0.84, 0.54]$$

$$E_{we} = [1.87 + 0, 1 + 0.09] = [1.87, 1.09]$$

$$E_{g0} = [-0.78 + 0.84, 0.54 + 1.27]$$

$$= [0.06, 0.81]$$

For we ,

$$r_{we} = [1.87, 1.09] \begin{vmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{vmatrix} = [-0.995, 3.7]$$

$$k_{we} = [1.87, 1.09] \begin{vmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{vmatrix} = [-4.8, 1.9]$$

$$v_{we} = [1.87, 1.09] \begin{vmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{vmatrix} = [2.5, -2.1]$$

For $g0$,

$$r_{g0} = [0.06, 0.81] \begin{vmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{vmatrix} = [-2.2, 1.98]$$

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$$K_{no} = \begin{bmatrix} 0.06 & 0.81 \end{bmatrix} \begin{vmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{vmatrix} = \begin{bmatrix} -1.3 & 0.75 \end{bmatrix}$$

$$V_{no} = \begin{bmatrix} 0.06 & 0.81 \end{bmatrix} \begin{vmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{vmatrix} = \begin{bmatrix} -0.153 & -0.222 \end{bmatrix}$$

Calculating score and softmax,

For we,

$$S (we \rightarrow we) = q_{we} \cdot k_{we} = 11.95$$

$$S (we \rightarrow no) = q_{we} \cdot k_{no} = 4.14$$

$$\text{Score}_{we} = \text{softmax}(11.95, 4.15)$$

$$= (0.99, 0.000408)$$

For no,

$$S (no \rightarrow we) = q_{no} \cdot k_w = 14.4$$

$$S (no \rightarrow no) = q_{no} \cdot k_{no} = 4.4$$

$$\text{Score}_{no} = \text{softmax}([14.4, 4.4])$$

$$= [0.99, 0.000046]$$

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Attention calculation

For w_e ,

$$\text{Score}_{w_e} = [0.99, 0.000408]$$

$$V_{w_e} = [2.5, -2.1]$$

$$V_{h_0} = [-0.153, -0.222]$$

$$0.99 \times V_{w_e} = [2.47, -2.08]$$

$$0.000408 \times V_{h_0} = [-0.0000624, -0.0000906]$$

(+) $\Rightarrow$

$$[2.47, -2.087]$$

For h_0 ,

$$\text{Score}_{h_0} = [0.99, 0.000046]$$

$$0.99 \times V_{w_e} = [2.47, -2.08]$$

$$0.000046 \times V_{h_0} = [-0.000007, -0.00001]$$

(+) $\Rightarrow$

$$[2.47, -2.08]$$

Final encoder embedding:

$$w_e = [1.87, 1.09] + [2.47, -2.087]$$

$$= [4.35, -1.00]$$

$$h_0 = [0.06, 0.81] + [2.47, -2.08]$$

$$= [2.54, -1.28]$$

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cross attention

$$Q = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix}$$

$$K = \begin{bmatrix} -1.1 & 0.3 \\ -1.5 & -0.8 \end{bmatrix}$$

$$V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & -0.5 \end{bmatrix}$$

$$EOS = [2.7, -1.34]$$

$$q_{EOS} = EOS \cdot Q = [3.648, 0.53]$$

$$k_{we} = we \cdot K = [-3.3, 2.1]$$

$$k_{h0} = h0 \cdot K = [-0.87, 1.78]$$

$$v_{we} = we \cdot V = [5.978, 3.11]$$

$$v_{h0} = h0 \cdot V = [4.33, 2.16]$$

Score + softmax:

$$S(EOS \rightarrow we) = q_{EOS} \cdot k_{we} = -10.87$$

$$S(EOS \rightarrow h0) = q_{EOS} \cdot k_{h0} = -2.2459$$

$$Score_{EOS} = \text{softmax}([-10.87, -2.2459])$$

$$= [0.000179, 0.999821]$$

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$$0.000179 \times v_{we} + 0.998 \times v_{q0} = [4.33, 2.16]$$

Final cross attention output:

$$\begin{aligned} E_{os-out} &= E_{os} [2.7, -1.34] + [4.33, 2.16] \\ &= [7.03, 0.82] \end{aligned}$$

Assume 2 inputs

to 33333333

33333333

33333333

Assume 2